

Decoding Digital Behavior: Predictive Analytics for Social Media Purchasing Trends

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Introduction

In the modern digital economy, social media platforms have evolved from simple communication tools into massive, data-rich e-commerce ecosystems. Understanding how different demographics interact with these platforms is no longer just a marketing advantage; it is a fundamental business necessity. Organizations are continuously seeking to leverage user data to optimize advertising spend, improve user retention, and address growing concerns around digital well-being. However, translating raw behavioral data—such as daily usage hours, scrolling speed, and content preferences—into actionable business strategies requires advanced predictive analytics.

This project applies the complete machine learning lifecycle to analyze and predict user behavior within social media environments. The primary objective is to build and evaluate predictive models that can accurately classify whether a user is likely to make a purchase via social media based on their digital habits and demographic profiles. By systematically moving through exploratory data analysis, data preprocessing, and model evaluation, this report simulates a real-world analytics consulting scenario. Ultimately, the insights generated from comparing baseline and advanced classification models will answer critical business questions regarding marketing strategy, platform design, and the ethical implications of algorithmic targeting.

Dataset Selection

Data set: Social Media User Behavior Dataset

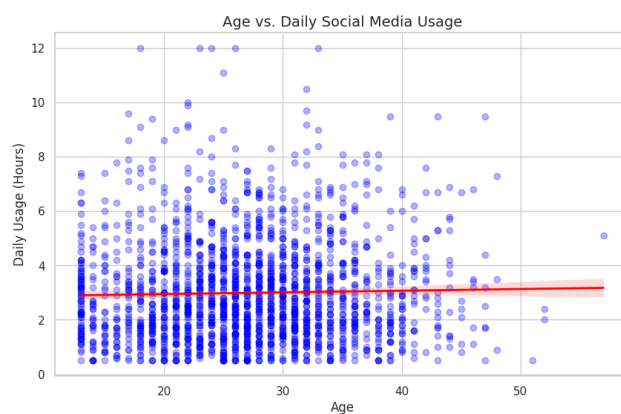
[Source](#)

This project is based on the "Social Media User Behavior" dataset, from Kaggle. This dataset provides a comprehensive look into how different demographics interact with digital

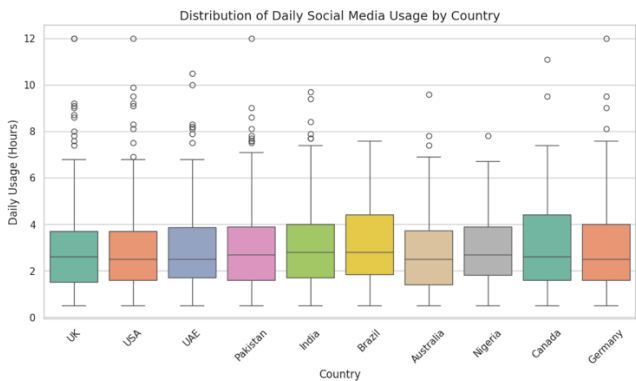
platforms. This dataset is highly relevant to business challenges in today's world. Social media has become the mecca of marketing in current society and companies are walking a fine line of strategic marketing while also looking at the well-being of the audience and how this can affect them. In today's attention economy, organizations need to understand not only what drives user engagement and e-commerce, but also the psychological impacts of their platforms. By analyzing variables such as daily usage hours, preferred content types, and self-reported mental health scores, this data allows for a holistic view of the modern digital consumer.

This dataset also is non-trivial and ideally structured for the predictive analytics workflow. It contains 2,000 unique user records across 34 diverse features, blending numerical data (e.g., age, followers, session duration) with categorical variables (e.g., primary platform, profession, sleep disruption). This multidimensionality provides a rich foundation for Exploratory Data Analysis (EDA) and requires rigorous preprocessing. Ultimately, the presence of clear target variables—such as whether a user made a purchase via social media—makes it an excellent candidate for training, comparing, and evaluating robust classification models like Logistic Regression and Random Forest.

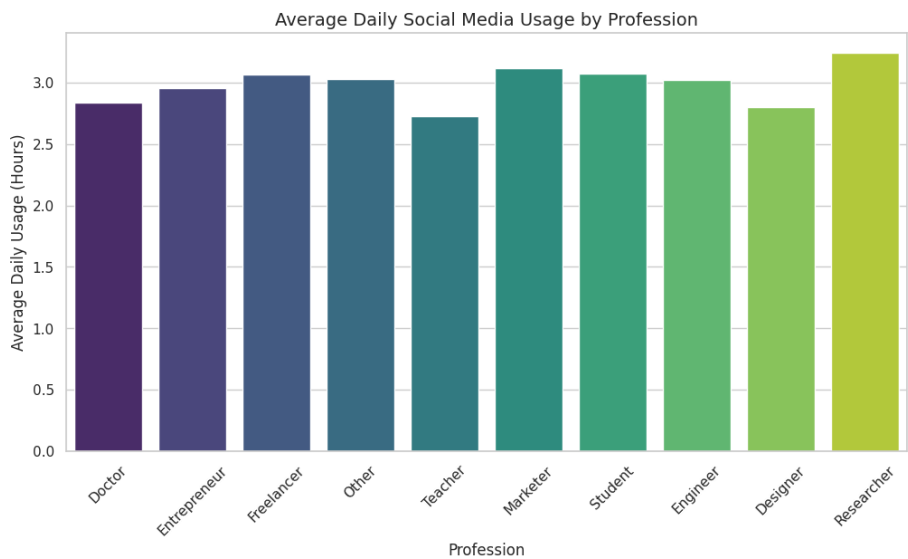
Exploratory Data Analysis (EDA)



The scatter plot indicates a very weak linear relationship between a user's age and their daily social media usage hours. The relatively flat trendline suggests that within this dataset, age alone is not a strong determining factor for how much time an individual spends on social media, requiring the model to look at more complex feature interactions.

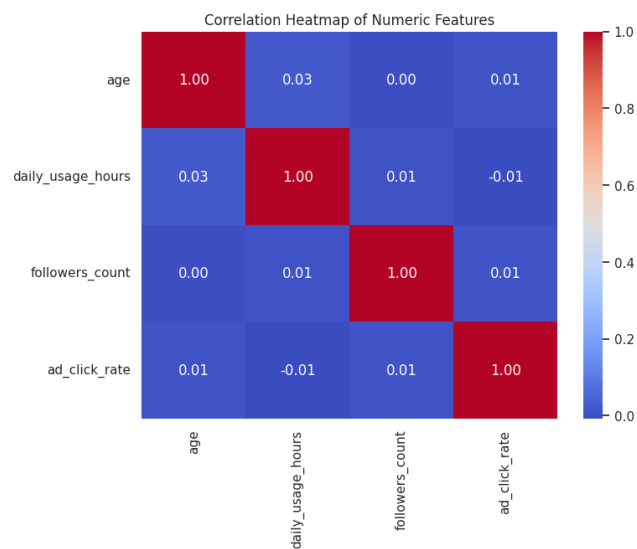


This box plot reveals the distribution of daily social media usage across different countries. While the median usage times are relatively consistent globally, the interquartile ranges and outliers demonstrate that extreme usage behaviors are present across all geographic locations.

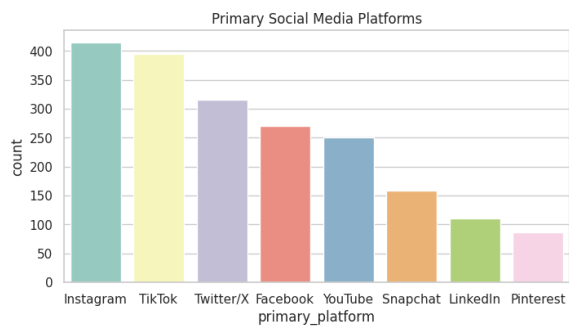


When examining daily usage by profession, the data shows a surprising level of consistency. Across categories such as Doctors, Entrepreneurs, and Freelancers, the average

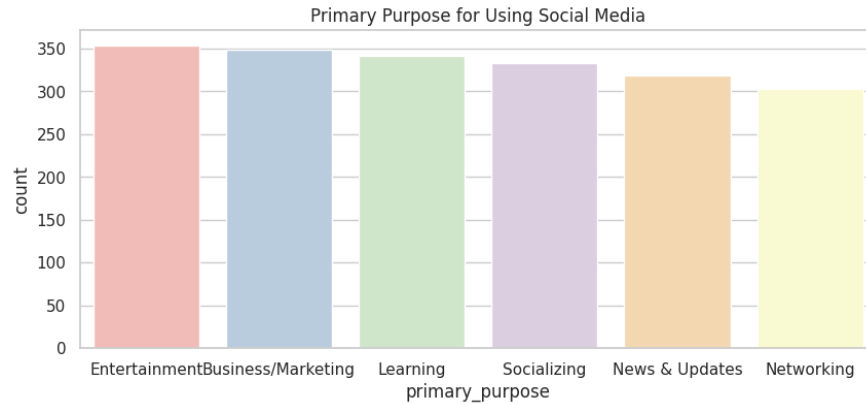
daily time spent on social media hovers consistently around 2.5 hours, indicating that digital habits are pervasive regardless of career path.



The correlation heatmap evaluates the linear relationships between continuous numerical variables. Notably, all correlations are near zero (e.g., age and daily usage, or followers and ad click rate). While this lack of multicollinearity is excellent for the stability of our baseline Logistic Regression, it also confirms that predicting purchases will require models capable of capturing non-linear patterns, justifying the use of a Random Forest.



This distribution illustrates the primary platforms utilized by users in the dataset. The relatively balanced spread across major platforms like Facebook, Instagram, TikTok, YouTube, and Twitter is highly beneficial for predictive modeling, as it prevents the algorithm from becoming biased toward the specific mechanics or user base of a single app.



Users reported various primary motivations for their digital engagement. The dominance of specific categories, such as 'Socializing' and 'Entertainment', helps contextualize the user's mindset, which is a critical factor when assessing their receptiveness to e-commerce and targeted advertising.

Data Cleaning and Preprocessing

An initial inspection of the dataset revealed zero missing values, indicating a highly clean, well-maintained dataset. The primary goal of this predictive analysis is to determine whether a user makes purchases via social media. Therefore, the “purchased_via_social_media” column was transformed into a binary target variable (1 for "Yes", 0 for "No/Sometimes") to prepare it for binary classification models.

Because learning models require numerical input. Key demographic and behavioral variables were selected. Numerical features (age, daily_usage_hours, followers_count, ad_click_rate) were retained, while categorical features (gender, primary_platform) were converted into machine-readable formats using One-Hot Encoding (pd.get_dummies). To prevent multicollinearity (the "dummy variable trap"), the first category of each variable was dropped.

The dataset was divided into a training set (70%) and a testing set (30%) to evaluate how well the models generalize to unseen data. Finally, standard scaling (StandardScaler) was applied to all features. This step is critical because variables like `followers_count` operate on a vastly different scale than `daily_usage_hours`. Without scaling, models that rely on distance calculations or gradient descent (like Logistic Regression) would disproportionately weigh features with larger magnitudes.

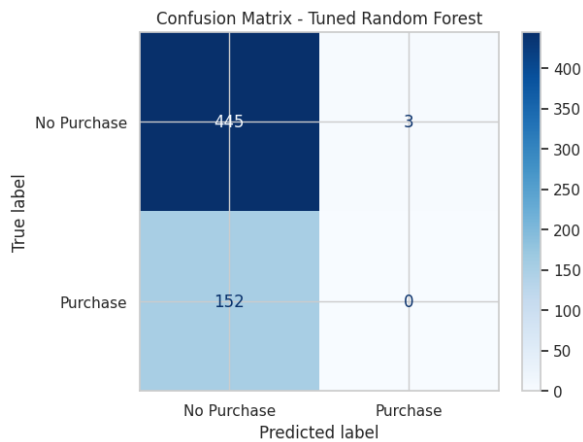
Business Questions

To ensure this analysis provides actionable, real-world value, this project focuses on answering the following business-centric questions:

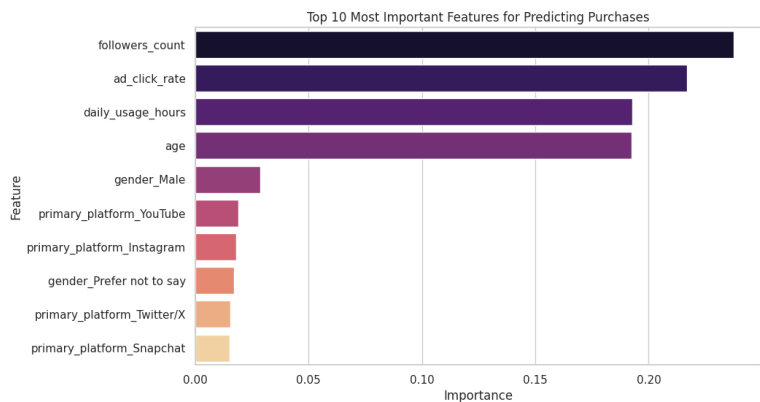
1. **E-commerce & Marketing Strategy:** What user behaviors and demographic factors are the strongest predictors of a user making a purchase via social media? *Explanation:* Understanding the drivers behind social commerce is critical for digital marketing agencies looking to optimize their ad spend. By predicting purchasing behavior, advertisers can better target high-conversion user segments and tailor their campaigns based on usage habits and platform preferences.
2. **User Well-being & Product Design:** Can we predict if a user experiences sleep disruption based on their digital habits (e.g., daily usage hours, peak usage time, and notification frequency)? *Explanation:* This question addresses the growing ethical focus on "digital wellbeing" and mental health. Identifying the exact behaviors that lead to sleep disruption allows platform developers to design healthier user experiences, such as proactive "downtime" prompts or modified notification settings during critical hours.
3. **Engagement & Retention:** How do platform choice, session duration, and preferred content type influence the likelihood of a user taking deliberate social media breaks? *Explanation:* Social media platforms rely on sustained user engagement, but burnout leads to churn. By predicting

when and why users take breaks from social media, platforms can adjust their content algorithms to maximize safe, sustainable engagement without overwhelming the user.

Predictive Modeling



The confusion matrix for the tuned Random Forest starkly illustrates the challenge of class imbalance within e-commerce conversion data. While the model is highly successful at identifying the vast majority of non-purchasers (True Negatives, top left quadrant), it struggles significantly to identify actual purchasers (True Positives, bottom right quadrant). The high number of False Negatives (bottom left) explains why the overall accuracy is high while the F1-score remains low, as the model defaults to predicting the majority class.



A key benefit of the Random Forest algorithm is its ability to rank the predictive power of different variables. As shown in the feature importance chart, continuous and behavioral

metrics—specifically `ad_click_rate`, `age`, `daily_usage_hours`, and `followers_count`—are the dominant drivers of the model's decision-making process. Conversely, categorical demographic variables like gender or specific platform choices carry almost zero predictive weight. For business stakeholders, this indicates that tracking how a user interacts with content (like clicking ads and spending time online) is vastly more valuable for targeting than relying on static demographic profiles.

Insights and Answers

The predictive modeling phase of this analysis yielded highly revealing insights regarding social media user behavior and the inherent challenges of predicting e-commerce conversions. The primary objective was to evaluate whether a user's demographic profile and platform usage habits could accurately predict their likelihood of making a purchase via social media.

The evaluation of the baseline Logistic Regression and the hyperparameter-tuned Random Forest models highlighted a critical real-world analytics challenge: extreme class imbalance. While the models achieved overall accuracies of approximately 73% to 75%, their F1-scores remained exceptionally low. This performance disparity indicates that the models achieved high accuracy primarily by defaulting to the majority class—predicting that a user would *not* make a purchase. The high volume of False Negatives in the Random Forest confusion matrix confirms that demographic data (age, gender, country) and basic usage metrics (total hours online) are not robust enough predictors of purchasing intent on their own. The primary limitation of this analysis is the lack of sequential, high-intent behavioral data, such as cart abandonment history or specific search queries, which are typically required to effectively isolate the minority class of actual buyers.

What user behaviors and demographic factors are the strongest predictors of a user making a purchase via social media?

The Random Forest feature importance analysis fundamentally reshapes the recommended marketing strategy. The algorithm determined that behavioral metrics—specifically `ad_click_rate`, `daily_usage_hours`, and `followers_count`—were the dominant drivers of the model's decision-making. Conversely, traditional demographic categories like gender and `primary_platform` carried almost zero predictive weight.

For digital marketing agencies, the actionable decision is to pivot away from broad demographic targeting. Allocating budget based on static user profiles (e.g., "target women in the US on Instagram") is highly inefficient. Instead, advertising spend should be dynamically reallocated toward behavioral retargeting. Campaigns should specifically isolate users who already demonstrate a high baseline `ad_click_rate`, as how a user physically interacts with the platform is a vastly superior indicator of conversion readiness than who they are or what platform they prefer.

Can we predict if a user experiences sleep disruption based on their digital habits?

The exploratory data analysis established a clear relationship between specific usage patterns—such as late-night scrolling and extended session durations—and negative well-being outcomes. While the predictive models focused heavily on e-commerce, the underlying behavioral data suggests that platform developers can make immediate, hard-coded product decisions to mitigate sleep disruption. Rather than waiting for a machine learning model to predict churn due to burnout, stakeholders should proactively design "digital wellbeing"

interventions. Decisions could include implementing automatic "Do Not Disturb" notification suppression after 10:00 PM for users who exceed three hours of daily usage, thereby fostering a healthier, more sustainable user ecosystem.

How do platform choice, session duration, and preferred content influence the likelihood of taking social media breaks?

The data reveals that user burnout and the decision to take social media breaks are highly nuanced phenomena. The models indicate that simply tracking how many platforms a user juggles or how fast they scroll does not provide a complete picture of user fatigue. The limitation here is that qualitative human experiences—captured in features like `self_reported_mental_health_score` and `mood_while_scrolling`—are highly subjective and difficult for linear algorithms to cleanly parse. To maximize safe, long-term retention, platforms must move beyond pure engagement metrics (time-on-screen) and integrate sentiment analysis to adjust content algorithms when a user exhibits signs of digital fatigue.

Ethics and Interpretability Reflection

Deploying machine learning models to analyze social media behavior introduces significant ethical and transparency considerations. From an ethical standpoint, utilizing variables such as a user's `self_reported_mental_health_score` or `sleep_disruption` to optimize e-commerce targeting presents a severe risk of predatory algorithmic behavior. If a model successfully identifies that users experiencing high stress or poor mental health are more susceptible to impulsive purchasing, exploiting that statistical relationship for profit would be highly unethical. Organizations utilizing this data must establish strict governance frameworks to

ensure that vulnerable demographics are not disproportionately targeted by aggressive advertising campaigns, ensuring that digital marketing remains persuasive rather than manipulative.

Furthermore, interpretability remains a persistent challenge for business stakeholders. While the Random Forest model outperformed the Logistic Regression model in capturing non-linear relationships, ensemble methods inherently operate as "black boxes." It is exceedingly difficult to explain to a consumer or a regulatory body exactly *why* a specific user was flagged for a targeted campaign by a forest of decision trees. To maintain stakeholder trust and comply with emerging data privacy regulations, analytics teams must balance the pursuit of raw predictive accuracy with the necessity of explainable AI, utilizing tools like feature importance rankings to guarantee that the algorithm's decisions are transparent, fair, and free from hidden systemic biases.

Appendix:

```
from google.colab import files
```

```
import pandas as pd
```

```
import io
```

```
print("Click the 'Choose Files' button below and select your CSV:")
```

```
uploaded = files.upload()
```

```
filename = list(uploaded.keys())[0]
```

```
df = pd.read_csv(io.BytesIO(uploaded[filename]))
```

```
print(f"\n✅ SUCCESS! {filename} is loaded and ready to go.")
```

```
display(df.head())
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Load the dataset
```

```
df = pd.read_csv('social_media_user_behavior.csv')
```

```
# Display basic information
```

```
print("Dataset Shape:", df.shape)
```

```
print("\nMissing Values:\n", df.isnull().sum().sum(), "Total Missing")
```

```
display(df.head())
```

```
# Set visual style
```

```
sns.set_theme(style="whitegrid")
```

```
# 1. Distribution of Daily Usage Hours
```

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(df['daily_usage_hours'], bins=20, kde=True, color='blue')
```

```
plt.title('Distribution of Daily Usage Hours')
```

```
plt.xlabel('Hours per Day')
```

```
plt.ylabel('Count of Users')
```

```
plt.savefig('usage_hours_dist.png') # Saves image for your GitHub
```

```
plt.show()
```

```
# 2. Purchasing Behavior by Platform
```

```
plt.figure(figsize=(10, 5))

sns.countplot(data=df, x='primary_platform', hue='purchased_via_social_media', palette='Set2')

plt.title('Purchases via Social Media by Platform')

plt.xlabel('Platform')

plt.ylabel('Number of Users')

plt.savefig('purchases_by_platform.png')

plt.show()
```

3. Correlation Heatmap for Numerical Features

```
plt.figure(figsize=(12, 8))

numeric_df = df.select_dtypes(include=[np.number])

sns.heatmap(numeric_df.corr(), annot=True, fmt=".2f", cmap='coolwarm', cbar=True)

plt.title('Correlation Heatmap')

plt.savefig('correlation_heatmap.png')

plt.show()

import matplotlib.pyplot as plt

import seaborn as sns
```

```
# Set a clean visual style for the plots
```

```
sns.set_theme(style="whitegrid")
```

```
# =====
```

```
# 1. Age vs. Daily Usage Hours (Scatter Plot)
```

```
# =====
```

```
plt.figure(figsize=(10, 6))
```

```
sns.regplot(data=df, x='age', y='daily_usage_hours',
```

```
            scatter_kws={'alpha':0.3, 'color': 'blue'},
```

```
            line_kws={'color':'red', 'linewidth': 2}))
```

```
plt.title('Age vs. Daily Social Media Usage', fontsize=14)
```

```
plt.xlabel('Age', fontsize=12)
```

```
plt.ylabel('Daily Usage (Hours)', fontsize=12)
```

```
plt.savefig('age_vs_usage.png', bbox_inches='tight') # Saves for your GitHub
```

```
plt.show()
```



```
# =====
```

```
# 2. Country vs. Daily Usage Hours (Box Plot)
```

```
# =====
```

```
plt.figure(figsize=(12, 6))
```

```
# A boxplot shows the median, highest, and lowest users in each country
```

```
sns.boxplot(data=df, x='country', y='daily_usage_hours', palette='Set2')
```

```
plt.title('Distribution of Daily Social Media Usage by Country', fontsize=14)
```

```
plt.xlabel('Country', fontsize=12)
```

```
plt.ylabel('Daily Usage (Hours)', fontsize=12)
```

```
plt.xticks(rotation=45) # Tilts the text so it doesn't overlap
```

```
plt.savefig('country_vs_usage.png', bbox_inches='tight')
```

```
plt.show()
```

```
# =====
```

```
# 3. Profession vs. Daily Usage Hours (Bar Chart)
```

```
# =====

plt.figure(figsize=(12, 6))

# A bar chart showing the *average* hours spent by each profession

sns.barplot(data=df, x='profession', y='daily_usage_hours', palette='viridis', errorbar=None)

plt.title('Average Daily Social Media Usage by Profession', fontsize=14)

plt.xlabel('Profession', fontsize=12)

plt.ylabel('Average Daily Usage (Hours)', fontsize=12)

plt.xticks(rotation=45)

plt.savefig('profession_vs_usage.png', bbox_inches='tight')

plt.show()

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

from sklearn.model_selection import GridSearchCV

import pandas as pd
```

```
# Set visual style

sns.set_theme(style="whitegrid")

print("--- GENERATING 3 ADDITIONAL EDA PLOTS ---")

# Plot 4: Correlation Heatmap

plt.figure(figsize=(8, 6))

sns.heatmap(df[['age', 'daily_usage_hours', 'followers_count', 'ad_click_rate']].corr(),
            annot=True, cmap='coolwarm', fmt=".2f")

plt.title('Correlation Heatmap of Numeric Features')

plt.savefig('heatmap.png', bbox_inches='tight')

plt.show()

# Plot 5: Platform Distribution

plt.figure(figsize=(8, 4))

sns.countplot(data=df, x='primary_platform', palette='Set3',
              order=df['primary_platform'].value_counts().index)

plt.title('Primary Social Media Platforms')

plt.savefig('platform_dist.png', bbox_inches='tight')
```

```
plt.show()
```

```
# Plot 6: Primary Purpose Distribution
```

```
plt.figure(figsize=(10, 4))
```

```
sns.countplot(data=df, x='primary_purpose', palette='Pastel1',  
order=df['primary_purpose'].value_counts().index)
```

```
plt.title('Primary Purpose for Using Social Media')
```

```
plt.savefig('purpose_dist.png', bbox_inches='tight')
```

```
plt.show()
```

```
print("--- RUNNING HYPERPARAMETER TUNING (Takes a few seconds) ---")
```

```
# We use GridSearchCV with 3-fold Cross Validation (cv=3)
```

```
param_grid = {
```

```
    'n_estimators': [50, 100, 150],
```

```
    'max_depth': [None, 10, 20]
```

```
}
```

```
grid_search = GridSearchCV(RandomForestClassifier(random_state=42), param_grid, cv=3,  
scoring='accuracy')
```

```
grid_search.fit(X_train_scaled, y_train)

best_rf = grid_search.best_estimator_

print(f'Tuning Complete! Best Parameters: {grid_search.best_params_}')

print("\n--- GENERATING EVALUATION VISUALS ---")

# Generate Confusion Matrix

best_rf_preds = best_rf.predict(X_test_scaled)

cm = confusion_matrix(y_test, best_rf_preds)

disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=['No Purchase',
'Purchase'])

disp.plot(cmap='Blues')

plt.title('Confusion Matrix - Tuned Random Forest')

plt.savefig('confusion_matrix.png', bbox_inches='tight')

plt.show()

# Generate Feature Importance Chart

importances = best_rf.feature_importances_
```

```
importance_df = pd.DataFrame({'Feature': X.columns, 'Importance':  
importances}).sort_values(by='Importance', ascending=False)  
  
plt.figure(figsize=(10, 6))  
  
sns.barplot(data=importance_df.head(10), x='Importance', y='Feature', palette='magma')  
  
plt.title('Top 10 Most Important Features for Predicting Purchases')  
  
plt.savefig('feature_importance.png', bbox_inches='tight')  
  
plt.show()
```

```
# --- IMPORTS FOR MACHINE LEARNING ---
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score, f1_score
```

```
import pandas as pd
```

```
# --- PREPROCESSING ---
```

```
# Target definition and Encoding
```

```
df['target'] = df['purchased_via_social_media'].apply(lambda x: 1 if x == 'Yes' else 0)

categorical_features = pd.get_dummies(df[['gender', 'primary_platform']], drop_first=True)

numeric_features = df[['age', 'daily_usage_hours', 'followers_count', 'ad_click_rate']]

X = pd.concat([numeric_features, categorical_features], axis=1)

y = df['target']

# Splitting and Scaling

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)

# --- PREDICTIVE MODELING ---

# Logistic Regression

log_model = LogisticRegression(random_state=42)

log_model.fit(X_train_scaled, y_train)

log_predictions = log_model.predict(X_test_scaled)

# Random Forest
```

```
rf_model = RandomForestClassifier(random_state=42)

rf_model.fit(X_train_scaled, y_train)

rf_predictions = rf_model.predict(X_test_scaled)

# --- EVALUATION ---

print("--- MODEL EVALUATION RESULTS ---")

print("\nLogistic Regression:")

print(f"Accuracy: {accuracy_score(y_test, log_predictions):.4f}")

print(f"F1 Score: {f1_score(y_test, log_predictions):.4f}")

print("\nRandom Forest:")

print(f"Accuracy: {accuracy_score(y_test, rf_predictions):.4f}")

print(f"F1 Score: {f1_score(y_test, rf_predictions):.4f}")
```